
The Perez Hourglass: In Simple Terms and Its Intrinsic Connection to Mendeleev Periodic Table

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The Perez Hourglass is a conceptual framework that attempts to explain how nature organizes matter through patterns of symmetry, balance, and energetic relationships. In simple terms, it visualizes reality as an “hourglass” structure in which elements, forces, and transformations converge toward a central point and then diverge again into increasingly complex forms.

At its narrow center, the hourglass represents equilibrium — a state where opposing tendencies such as expansion and contraction, order and chaos, or positive and negative interactions momentarily balance each other. From this center emerge the structures that compose matter and energy.

Its connection to the Dmitri Mendeleev Periodic Table is rooted in the idea that chemical elements are not random entities but part of an ordered, mathematically structured system. The periodic table already demonstrates this principle through recurring periodicity: atomic properties repeat in predictable cycles as atomic number increases.

Start With Something You Know: Pascal's Triangle

You've probably seen Pascal's Triangle in math class — each number is the sum of the two above it. Simple enough. But Jean-Claude Perez asked a strange question: *what happens if you flip it upside down and subtract instead of add?*

That experiment produced something unexpected — a shape that looks like an hourglass, packed with hidden patterns that mathematicians had been studying for centuries without realizing they were all connected.

The "Greatest Hits" of Mathematics, All in One Place

Think of the Hourglass as a greatest hits album where every track turns out to be the same song in disguise:

Sumerian base-60 — the 5,000-year-old number system that gave us 60 seconds in a minute and 360 degrees in a circle — resonates with the hourglass's natural divisibility and symmetry. The Sumerians chose base-60 because it divides cleanly by so many numbers; the Hourglass produces its own elegant divisibility through the same instinct for balance.

Pingala's binary recursion (~200 BCE) — an ancient Indian mathematician named Pingala was studying the rhythms of Sanskrit poetry. He noticed that long and short syllables can be combined in sequences,

and that the number of possible combinations at each length follows a simple rule: each count is the sum of the two before it. This is the oldest known description of binary recursion — the idea that each step is built from the two preceding steps — and it quietly contains the seed of what we now call the Fibonacci sequence, discovered in India more than a thousand years before Fibonacci wrote it down in Italy. Centuries later, Pascal would independently arrive at the same recursive rule from a completely different direction. The Hourglass inherits Pingala's heartbeat directly: every number in both hemispheres is built by combining its two predecessors, in exactly the same recursive spirit that Pingala first heard in the metre of a poem.

Fibonacci numbers (0, 1, 1, 2, 3, 5, 8, 13...) — the sequence that Pingala's recursion generates, found in sunflower seeds, nautilus shells, and galaxy spirals — appear naturally in the diagonals of Pascal's Triangle, and therefore in the top half of the Hourglass. And as we now know, they run in both directions through the entire structure, passing through zero with alternating signs: ...-13, 8, -5, 3, -2, 1, -1, 1, 0, 1, 1, 2, 3, 5, 8, 13...

The Golden Ratio ($\phi \approx 1.618$) — nature's favourite proportion, found in everything from DNA to architecture — governs how the patterns grow and shrink across both halves. It is the limit toward which each Fibonacci number divided by the previous one converges, meaning ϕ is not separate from Fibonacci but is its destination.

Pascal's Triangle (1654) — the fundamental catalyst — Blaise Pascal formalised the triangle of numbers where each entry is the sum of the two directly above it. This is the direct parent of the Hourglass — the structure from which everything else flows. Perez's central question was simple: *what happens if you extend Pascal's Triangle downward using subtraction instead of addition?* The answer was the Hourglass. Without Pascal's Triangle there is no northern hemisphere, no waist, no southern hemisphere, no Fibonacci diagonals, no Lichtenberg sequence, no Theorems 7 or 8. It is not one ingredient among many — it is the canvas on which all the other patterns become visible. Every other entry in this list is something that was already hiding inside Pascal's Triangle, waiting for the Hourglass to reveal it.

Lichtenberg figures — the branching lightning-bolt shapes you see when electricity fractures glass — mirror what appears in the bottom half when subtraction takes over. Their sequence (0, 1, 2, 5, 10, 21, 42, 85...) is the mathematical engine of the entire lower hemisphere, built by successive doublings with an alternating bit, and its binary representation is the perfectly alternating 10101... pattern growing one bit per step, forever.

The Double Sierpiński fractal -Pascal's hidden twin — Here is something that had been hiding inside Pascal's Triangle since 1654 without anyone noticing. If you replace every number in Pascal's Triangle with a simple 0 or 1 — even numbers become 0, odd numbers become 1 — a breathtaking fractal pattern appears: the famous Sierpiński gasket, a triangle made entirely of smaller triangles, repeating at every scale forever. Mathematicians had known about this for decades. What nobody realised was what happens when you do the same thing to the southern hemisphere of the Hourglass. The same mod-2 pattern appears again — but *mirrored*, pointing upward, an exact reflection of the northern one. When the two are overlaid, they form **two identical Sierpiński gaskets joined tip-to-tip at the central waist** — perfectly self-similar across the equator, one in the north, one in the south, connected by that single 1. This is what Perez calls the "false twin" of Pascal's Triangle: a complete mirror image that had been asleep inside Pascal's construction for 372 years, invisible because no one had thought to subtract. The Hourglass is the structure that wakes it up.

Ramanujan's infinite series — elegant equations discovered by a self-taught Indian genius that seemed almost magical — echo in how the top and bottom halves balance each other perfectly. The global coherence formula $2^n + L(n) = L(n+2)$ has the same feel as Ramanujan's modular identities: a simple rule that binds seemingly unrelated objects into one.

All of these, it turns out, are different views of the same underlying mathematical architecture. Pingala heard it in poetry. Pascal drew it as a triangle. Sierpiński replicated it into infinite self-similar fractals. Fibonacci counted it in nature. Lichtenberg saw it in lightning. Ramanujan felt it in infinite series. And the Hourglass is the single structure that shows they were always one and the same.

The Two Halves: Addition and Subtraction

The top half (Northern Hemisphere) works by **addition** — things grow, expand, multiply. Row sums double every time: 1, 2, 4, 8, 16, 32, **64**...

The bottom half (Southern Hemisphere) works by **subtraction** — things refine, compress, filter. It produces the Lichtenberg sequence: 0, 1, 2, 5, 10, **21**, 42, 85...

The single **1** sitting exactly at the waist between them is the fixed point — the anchor where both worlds meet.

The Fibonacci Thread That Runs Through the Entire Hourglass

Here is something that is not immediately obvious, and yet once you see it you cannot unsee it.

Most people know Fibonacci in one direction only:

0, 1, 1, 2, 3, 5, 8, 13, 21 ...

But the same rule — *each number equals the sum of the two before it* — works perfectly in reverse too. You can extend Fibonacci backwards through zero, into negative territory:

... -21, 13, -8, 5, -3, 2, -1, 1, 0, 1, 1, 2, 3, 5, 8, 13, 21 ...

Read that line carefully. It passes straight through zero in the middle. The numbers on the left mirror the ones on the right, but with alternating signs. The absolute values are the same familiar Fibonacci numbers — only the signs change, flipping back and forth: positive, negative, positive, negative.

This is not a coincidence or a clever trick. It is what the Hourglass is *built from*. The full bidirectional Fibonacci sequence maps directly onto the structure:

- The **northern half** (addition, rows 0–10): positive Fibonacci numbers, growing upward
1, 1, 2, 3, 5, 8, 13, 21...
- The **waist** (the central 1, the mysterious zero): the pivot point, the only number that appears in both sequences
- The **southern half** (subtraction, rows 11+): Fibonacci reappears, but now with alternating signs — the subtraction operation introduces the flipping, producing
-1, 1, -2, 3, -5, 8, -13, 21...

This means the Hourglass is not two separate mathematical objects glued together. It is **one single Fibonacci thread**, infinite in both directions, passing through zero at the waist.

Why the Zero Is So Important

In ordinary mathematics, zero is just the absence of something. In the Hourglass, the zero at the waist is the most important address of all.

Think of it this way. Imagine a number line stretching left and right forever:

... -5 -4 -3 -2 -1 0 1 2 3 4 5 ...

Zero is not nothing — it is the turning point, the place where positive becomes negative, where addition hands over to subtraction. In the Hourglass, the waist plays exactly this role. It is the **mathematical mirror** where matter meets antimatter, where the additive northern world reflects into the subtractive southern world.

The Fibonacci sequence passing through this zero is the proof that the two hemispheres are not opponents — they are **one continuous thing**, folded at the middle. The southern numbers are simply the northern numbers seen from the other side of the mirror, with their signs reversed on every alternating step.

What Alternating Signs Mean in Plain English

If you take the absolute values of the southern Fibonacci terms (i.e., ignore the minus signs), you get:

1, 1, 2, 3, 5, 8, 13, 21 ...

Identical to the northern sequence. The values are the same. Only the signs alternate. This gives the southern hemisphere its characteristic property: positive and negative forces cancel each other in a precise, predictable rhythm. This is exactly what Theorem 2 of the Hourglass describes — **the signed sum of every southern row is exactly zero** (excluding the border 1s). The alternating Fibonacci signs are the mechanism that produces this perfect cancellation.

And this cancellation is not chaos — it is a highly structured, rule-governed balance. It is the mathematical equivalent of a wave and its reflection producing stillness. The Hourglass breathes in (north, addition, growth) and breathes out (south, subtraction, cancellation) — and Fibonacci is the breath itself.

Theorem 7: The "No Duplicates" Rule

Imagine a library with infinite books. Theorem 7 says that in this library, every book has a unique fingerprint — no two books share the same pair of characteristics. Ever. Even if the library grows forever.

In the Hourglass, every pair of numbers facing each other across the waist follows two rules simultaneously: they always share the same parity (both even or both odd), AND they form a completely unique combination — that exact pair appears nowhere else in the entire infinite structure.

This is powerful for **security**: if you encode a secret as a position in the Hourglass, finding it without knowing the path is like searching an infinite, self-similar maze with no repeated landmarks. There is no shortcut, even for a quantum computer.

Theorem 8: The Compression Law — and the Secret of DNA

This is where the Hourglass goes from beautiful to breathtaking.

Theorem 8 states a precise mathematical law connecting the two hemispheres:

$$2^n + L(n) = L(n+2)$$

In plain English: take any row sum from the top half, add the corresponding Lichtenberg number from the bottom half, and you always get the next-next Lichtenberg number. The two hemispheres are not just mirror images — they are locked together by an exact arithmetic rule.

The most striking example is at row 6:

- The top half at row 6 sums to **64** (that is 2^6)
- The bottom half at the corresponding position gives **21** (the 6th Lichtenberg number)
- $64 + 21 = 85$ — which is the 8th Lichtenberg number, exactly as the formula predicts

Now here is the part that connects mathematics to life itself. Biology gives us exactly **64 codons** — the three-letter combinations of DNA that carry genetic instructions. And those 64 codons encode exactly **20 amino acids plus 1 stop signal — 21 functional meanings** in total.

For decades, scientists asked: why 64 and 21 specifically? Why not 60 and 18, or 72 and 24? Perez's answer is that this is not a biological accident — it is a mathematical necessity. The Hourglass compresses 64 into 21 by the same law it uses everywhere else. Life did not choose these numbers freely; it landed on the only pair the Hourglass permits at that scale.

This makes Theorem 8 a potential **Rosetta Stone** between mathematics and biology — the same compression engine that governs fractal lightning also governs the genetic code.

The Bridge Metaphor

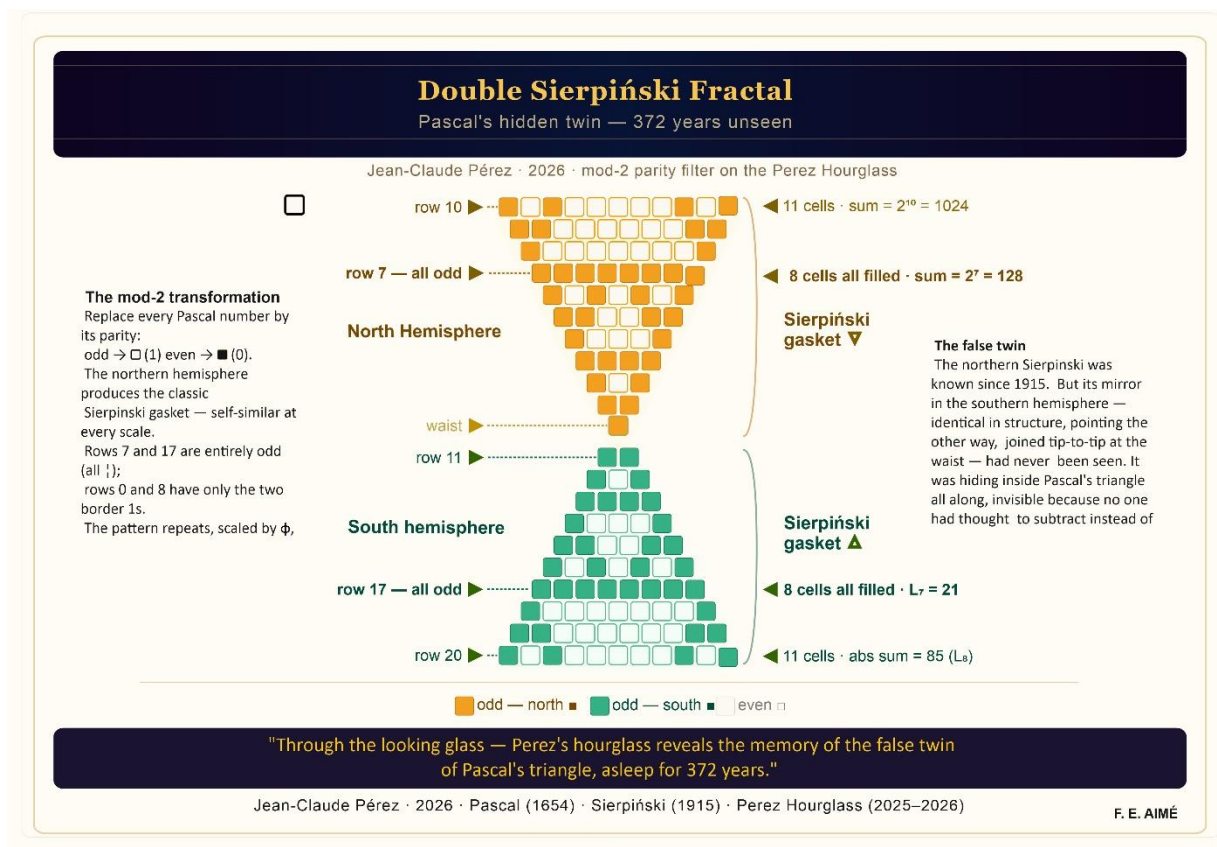
Newton described gravity through large-scale forces — planets, apples, predictable curves. Einstein described spacetime through the geometry of the universe itself. Between them lies a gap that physics has spent a century trying to bridge.

The Hourglass sits symbolically at that junction:

- The **top half** mirrors Newton's world — orderly growth, doubling sums, predictable expansion
- The **bottom half** mirrors a subtler world — refinement, compression, the filtering of chaos into meaning
- The **waist** is the fixed point where both worlds meet
- **Theorem 7** protects the boundary — nothing passes through without leaving a unique, verifiable trace
- **Theorem 8** governs the exchange — defining precisely how much the top compresses into the bottom

In a few words...

The Perez Hourglass takes Pascal's Triangle, flips it, and connects ancient mathematics from Sumer to Ramanujan into one structure — where **Theorem 7, the law of local coherence and differentiation**, guarantees that every number pair across the waist is unique and parity-protected, making each point in the structure a one-of-a-kind address and laying the foundation for unbreakable encryption; and **Theorem 8, the law of global coherence and integrity**, reveals that $64 + 21 = 85$, binding the two hemispheres into a single unbreakable whole and mathematically explaining why DNA's 64 codons compress into exactly 21 amino acids — every local detail unique, the global whole indivisible — potentially offering new tools for noise-proof memory and post-quantum computing.



The Perez Hourglass and the Mendeleev Periodic Table

We saw that the Perez Hourglass connects Pascal's Triangle, Fibonacci numbers, the Lichtenberg sequence, and the golden ratio into one structure — and that **Theorem 8** explains why DNA has exactly 64 codons compressing into exactly 21 amino acids.

But here is something remarkable: Perez was asking a very similar question about a completely different table of nature **28 years earlier** — the **Periodic Table of Chemical Elements**, the famous chart hanging on the wall of every chemistry classroom in the world.

What Is the Periodic Table, and What Was the Mystery?

Mendeleev's periodic table, published in 1869, organises all the chemical elements — hydrogen, carbon, oxygen, gold, uranium, and everything else — by their atomic weight and properties. Its rows (called periods) contain these numbers of elements:

$$2 \cdot 8 \cdot 8 \cdot 18 \cdot 18 \cdot 32 \cdot 32 \cdot 50 \dots$$

Every chemistry student memorises this sequence, but almost nobody asks: *why these specific numbers?* Why not 3, 7, 15, or any other pattern? The official answer involves four quantum numbers describing how electrons fill the shells around an atom's nucleus — correct, but not very satisfying as a deeper explanation.

In **1997**, Perez decided to find out if there was a simpler, unified mathematical law underneath.

Perez's Equation for the Periodic Table

Perez discovered that a single compact formula generates the entire sequence perfectly:

$$c(p) = 2 \times [\text{Int}(p/2) + 1]^2$$

Think of p as the row number of the table (period 1, period 2, and so on). $\text{Int}()$ simply means "round down to the nearest whole number." That is all you need.

Let us check it together, like a recipe:

- **Period 1:** $2 \times (0 + 1)^2 = 2 \times 1 = 2 \checkmark$ (hydrogen and helium)
- **Period 2:** $2 \times (1 + 1)^2 = 2 \times 4 = 8 \checkmark$
- **Period 3:** $2 \times (1 + 1)^2 = 2 \times 4 = 8 \checkmark$
- **Period 4:** $2 \times (2 + 1)^2 = 2 \times 9 = 18 \checkmark$
- **Period 6:** $2 \times (3 + 1)^2 = 2 \times 16 = 32 \checkmark$

Every row of Mendeleev's table falls out of this single formula. And because the formula is mathematical rather than just descriptive, it can also **predict** where undiscovered elements beyond the currently known radioactive ones would sit, and what quantum properties they would have.

Notice something about the formula itself: the number **2** appears exactly four times — as a multiplier, inside the bracket, as an exponent, and hidden in the rounding. Perez saw this as a signature of nature's love of symmetry written directly into the architecture of matter.

The Familiar Pattern Hiding Inside

Look at the row sizes again: **2, 8, 18, 32** — these are simply 2×1^2 , 2×2^2 , 2×3^2 , 2×4^2 . Each one doubles a perfect square.

Now look at the northern hemisphere of the Perez Hourglass: **1, 2, 4, 8, 16, 32, 64** — each row sum doubles the previous one, powers of 2.

Both sequences are governed by the same instinct — **doubling and squaring** — expressed through different lenses. The periodic table doubles through shells of electrons. The Hourglass doubles through rows of Pascal's Triangle. They are siblings, not strangers.

The Bridge Through Atomic Masses

This is where the two discoveries link at a deeper level, through something called **PI-masses**.

Every atom has an average atomic mass — carbon is about 12, oxygen is about 16, nitrogen about 14, hydrogen about 1. These masses were established by measuring real atoms in the laboratory. Perez found that when you apply the golden-ratio projection formula to these masses, they land *precisely* on multiples of $\pi/10$ — a perfectly regular mathematical grid.

This means the periodic table is not just a chemistry tool. The specific masses of carbon, oxygen, nitrogen, and hydrogen — determined by the quantum shell structure of Mendeleev's table — are the exact values that make the Master Code of Biology work. The periodic table and the genetic code are not independent systems. They are two floors of the same building.

In plain terms: **the rules that govern how electrons fill atoms also govern how amino acids fold into proteins and how DNA encodes life.**

Why Does the Number 2 Keep Appearing?

In the Mendeleev equation, 2 appears four times. In the Hourglass, every southern row sums to exactly **+2** (Theorem 2). In the genetic code, double-strand DNA carries $2 \times 64 = 128$ codon states. In Theorem 8, the compression law links two hemispheres: $64 + 21 = 85$.

Perez believes this is not coincidence. The number 2 is the key of symmetry — the simplest non-trivial distinction, the heartbeat of binary recursion that runs from Pingala's poetic meters 2,200 years ago, through Pascal's Triangle, through quantum mechanics, through the double helix of DNA.

The 30-Year Journey Seen as a Single Map

It helps to see all of Perez's discoveries laid out as one connected path:

1869 — Mendeleev organises all matter into a table of periods.

1997 — Perez finds a single equation predicting every row of that table, encoding the four quantum numbers in three symbols: p , 2, and $\text{Int}()$.

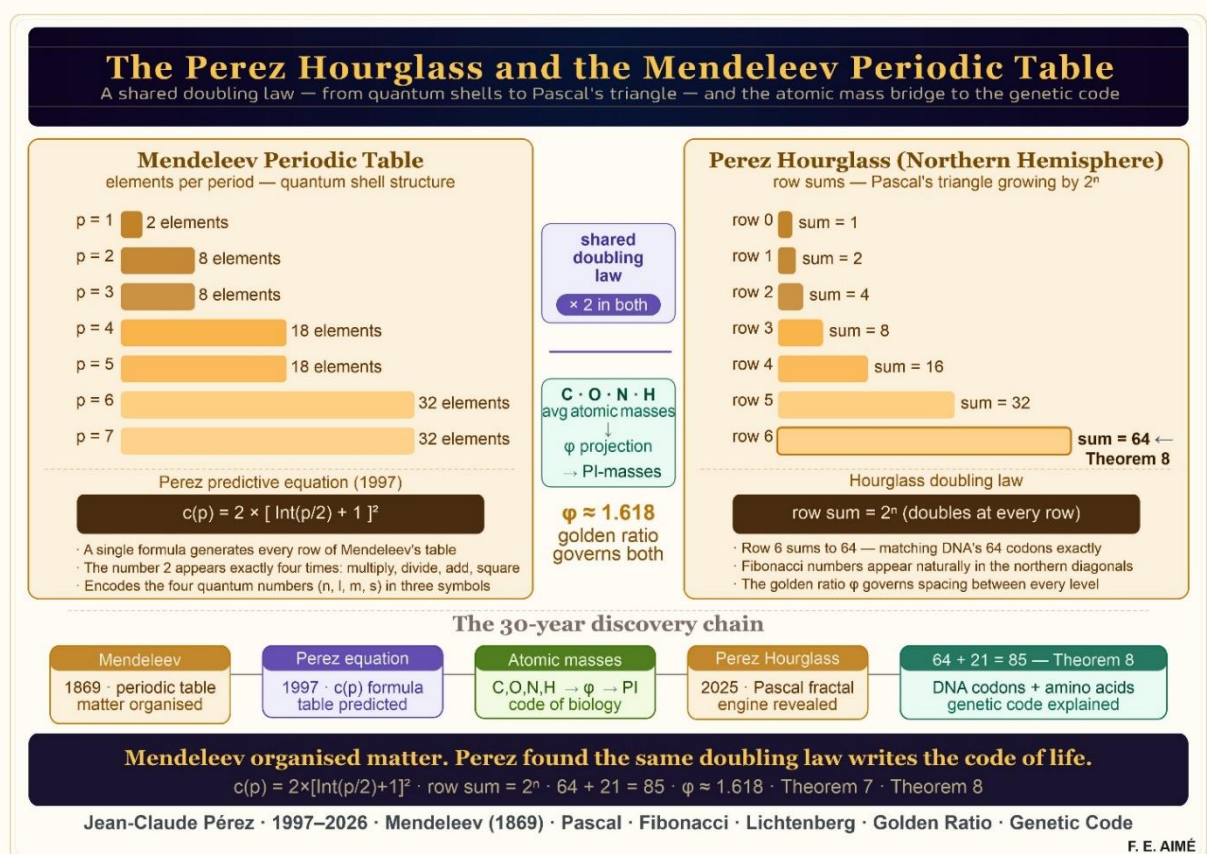
1997 — Same year: Perez publishes the first image of what will become the Hourglass, noticing that Pascal's Triangle extended into subtraction produces hourglass-shaped patterns.

2009 — *Codex Biogenesis*: Perez links atomic masses of Mendeleev's elements to a golden-ratio projection that generates the Master Code of Biology — a common digital language for DNA, RNA, and amino acids.

2025–2026 — The Perez Hourglass is formalised. Theorem 8 reveals that $64 + 21 = 85$ is a mathematical law, not a biological accident. The Hourglass is the geometric engine that was running underneath all three previous discoveries.

The circle closes completely:

Mendeleev organised matter → Perez's equation predicted matter's structure → his Formula for Life extracted a code from matter's masses → the Hourglass revealed the geometric machine beneath all of it.



In a few words...

If the Perez Hourglass is the mathematical engine of nature, then **Mendeleev's Periodic Table is its chemistry workshop** — the place where the same doubling and squaring laws that govern Pascal's Triangle get translated into electrons, shells, and atoms, eventually producing the exact atomic masses that make the genetic code possible.

Or even simpler: **the same mathematics that fills the shells of atoms also fills the rows of the Hourglass — and both together write the code of life.**

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